Sri Chaitanya ACADEMY | Infinity Learn



JEE CHAMPION TEST SERIES

JEE Main Full Syllabus Test - 7

Test Date: 19th Mar 2025

INSTRUCTIONS

- The test is of 3 hours duration.
- The Test Booklet consists of 75 questions. The maximum marks are 300.
- There are three parts in the question paper (1, 2, 3) consisting of Physics, Chemistry and Mathematics, having 25 questions in each subject and each subject having two sections.
  (I) Section-A contains 20 multiple choice questions with only one correct option.
  (II) Section-B contains 5 numerical value type questions.
- Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.
- The answer should be within 0 to 9999. If the answer is in decimal, then round off to the nearest integer value (e.g., If the answer is above 10 and less than 10.5, round off to 10; if the answer is more than or equal to 10.5 but less than 11, round off to 11).
- Use blue/black ballpoint pen only for writing particulars or marking responses on the answer sheet. Use of pencil is strictly prohibited.

MATHEMATICS

Max Marks: 100

SECTION-I (SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

1. The shortest distance between the lines $x = y + 2 = 6z - 6$ and $x + 1 = 2y = -12z$ is

1) $\frac{1}{2}$ **2)** 2 **3)** 1 **4)** $\frac{3}{2}$

2. Let 'α' and $f(\alpha)$ be the eccentricity of the ellipse $\frac{x^2}{4b^2 - 3a^2} + \frac{y^2}{3b^2 - 2a^2} = 1$ and

$\frac{x^2}{3b^2 - 2a^2} + \frac{y^2}{b^2} = 1$ respectively, where $b^2 > a^2 > 0$, then

1) $\underbrace{f\left(f\left(f.....f(\alpha)\right)\right)}_{2025\ times} = \frac{2^{\frac{2025}{2}}\alpha}{\sqrt{\left(1+\alpha^2\right) - 2^{2025}\alpha^2}}$

2) $\underbrace{f\left(f\left(f.....f(\alpha)\right)\right)}_{2025\ times} = \frac{2^{2025}\alpha}{\sqrt{1 - 2^{2025}\alpha^2}}$

3) $\underbrace{f\left(f\left(f\left(f.....f(\alpha)\right)\right)\right)}_{2025\ times} = \frac{2^{\frac{2025}{2}}\alpha}{\sqrt{1-(2024)\alpha}}$

4) $\underbrace{f\left(f\left(f.....(f(\alpha))\right)\right)}_{2024\ times} = \frac{2^{2024}\alpha}{\sqrt{\left(1+\alpha^2\right) - 2^{2024}\alpha^2}}$

3. If $x_i \in [0,1] \forall i = 1,2,3,.....28,$ then the maximum value of

$\sqrt{\sin^{-1}x_1}\sqrt{\cos^{-1}x_2} + \sqrt{\sin^{-1}x_2}\sqrt{\cos^{-1}x_3} + \sqrt{\sin^{-1}x_3}\sqrt{\cos^{-1}x_4} + ... + \sqrt{\sin^{-1}x_{28}}\sqrt{\cos^{-1}x_1}$ is

1) 7π **2)** 14π **3)** 28π **4)** can't be determined

4. Let R be the real line. Consider the following subsets of the plane $R \times R$:

$S = \{(x,y) : y = x + 1 and\ 0 < x < 2\}$

$T = \{(x,y) : x - y \text{ is an integer}\}$

Which one of the following is true

1) T is an equivalence relation on R but S is not

2) Neither S nor T is an equivalence relation on R

3) Both S and T are equivalence relations on R

4) S is an equivalence relation on R but T is not

5. If 'A' and 'B' are square Matrices of order '2' such that $A+adj\left(B^T\right)=\begin{bmatrix}2 & 1\\ 1 & 2\end{bmatrix}$ and $A^T-adj\left(B\right)=\begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}$, then

1) $A^n=A, n\in\mathbb{N}$ **2)** B is Skew symmetric matrix

3) $\left|A+A^2+A^3+A^4+A^5\right|=0$ **4)** $\left|B+B^2+B^3+B^4+B^5\right|=0$

Note: here $\left|K\right|$ denotes determinant of matrix K.

6. Five points are selected on a circle of radius 'r' the centers of the Rectangular Hyperbolas each passing through four of these points lie on a circle of radius

1) r **2)** 2r **3)** $\frac{r}{2}$ **4)** $\frac{r}{\sqrt{2}}$

7. The particular solution of Differential equation $x^2dy-y^2dx+xy^2\left(x-y\right)dy=0$ at $x=1\ and\ f\left(1\right)=2$ is

1) $\frac{y-x}{xy}=\frac{1}{2}e^{\frac{y^2-4}{2}}$ **2)** $2\left(\frac{y-x}{xy}\right)=\frac{5}{2}e^{\frac{y^2}{2}-2x}$

3) $\frac{y-x}{xy}=\frac{2}{5}e^{2+2y}$ **4)** $\frac{y-x}{xy}=e^{\frac{y^2}{2}-x}$

8. The sum of series $\sum_{r=0}^{n}(-1)^r\ {}^nC_r\left(\frac{1}{2^r}+\frac{3^r}{2^{2r}}+\frac{7^r}{2^{3r}}+\frac{15^r}{2^{4r}}+....+m\,termes\right)$ is

1) $\frac{2^{mn}-1}{2^{mn}\left(2^n-1\right)}$ **2)** $\frac{2^{mn}-1}{2^n-1}$ **3)** $\frac{2^{mn}+1}{2^n+1}$ **4)** $\frac{2^m}{2^n+1}$

9. If $h\left(x\right)=\underset{n\to\infty}{Lt}\frac{x^{2n}f\left(x\right)+g\left(x\right)}{x^{2n}+1}$, then the value of $h\left(x\right)$ when $x\in\mathbb{R}-\left[-1,1\right]$ is

1) $g\left(x\right)$ **2)** $f\left(x\right)$ **3)** $\frac{f\left(x\right)+g\left(x\right)}{2}$ **4)** $\frac{f\left(x\right)-g\left(x\right)}{2}$

10. The sum of the terms of the G.P $a,ar,ar^2,....\infty$ where 'a' is the only integer solution for the inequality $\log_{\frac{1}{2}}\left(\frac{2a-5}{a-3}\right)\geq 0$ and 'r' is given by the relation $\frac{1}{r}=e^{\left(\int_0^{\sqrt{2}-1}\frac{1}{1+x}dx\right)}$ is

1) $\frac{2}{\sqrt{2}-1}$ **2)** $\frac{\sqrt{2}}{\sqrt{2}-1}$ **3)** $\frac{2\sqrt{2}}{\sqrt{2}+1}$ **4)** $\frac{2\sqrt{2}}{\sqrt{2}-1}$

11. A vector $\bar{x}$ coplanar with vectors $\bar{a}=-\hat{i}+\hat{j}+\hat{k}$ and $\bar{b}=2\hat{i}+\hat{k}$ and is orthogonal to the vector $\bar{b}$. If $\bar{x}.\bar{a}=7$, Then the vector $\bar{x}$ equals____

1) $-\frac{3}{2}\hat{i}+\frac{5}{2}\hat{j}+3\hat{k}$ **2)** $\frac{3}{2}\hat{i}+\frac{5}{2}\hat{j}+3\hat{k}$ **3)** $3\hat{i}-5\hat{j}+\hat{k}$ **4)** $\frac{7}{2}\hat{i}-\frac{5}{2}\hat{j}+\hat{k}$

12. 2n boys are randomly divided into two groups each containing n boys. The chance that two particular boys X and Y are different groups is

1) $\frac{3n}{4n-1}$ **2)** $\frac{n-1}{2n-1}$ **3)** $\frac{n}{2n-1}$ **4)** $\frac{n+1}{2n+1}$

13. Statement-I: A fair coin is tossed n times. Let P_n denotes the probability that no two (or more) consecutive heads occur in n tosses. The probabilities P_2,P_3,P_4 are in arithmetic progression.

Statement-II: The probabilities $P_1,P_2,P_3,.....,P_n$ are in decreasing order

1) Both Statement-I and Statement-II are true.
2) Statement-I is true but Statement-II is false
3) Statement-I is false but Statement-II is true
4) Both Statement-I and Statement-II are false

14. The number of all possible values of θ, where $0<\theta<\pi$, for which system of questions $(y+z)\cos 3\theta=(xyz)\sin 3\theta$;

$x\sin 3\theta=\frac{2\cos 3\theta}{y}+\frac{2\sin 3\theta}{z}$;

$(xyz)\sin 3\theta=(y+2z)\cos 3\theta+y\sin 3\theta$ have a solution (x_0,y_0,z_0) with $y_0z_0\neq 0$, is____.

1) 2 **2) 4** **3) 3** **4) 5**

15. The value of $\frac{\sum_{r=0}^{n-1}\pi^r}{\sum_{r=0}^{n}\pi^r}-\frac{\sum_{r=1}^{n}e^{r-1}}{\sum_{r=1}^{n+1}e^{r-1}}$ is

1) >0 **2)** <0 **3)** $=0$ **4)** ≤ 0

16. If vector $\vec{r}$ with d.c.s. *l*, *m*, *n* is equally inclined to the co-ordinate axes, then the total number of such vectors is

1) 4 **2)** 6 **3)** 8 **4)** 2

17. If $z+\frac{1}{z}=\sqrt{3}$, then the value of $\sum_{r=1}^{5}\left(z^r+\frac{1}{z^r}\right)^2$ is_____.

1) 8 **2) 5** **3) 4** **4) 1**

18. Match the following columns and select the correct option

List – I		List – II	
P	The maximum value of $f(x)=9x(x-1)^2, 0\leq x\leq 2$ is	A	-1

Q	The minimum value of $f(x)=\frac{x^2-1}{x^2+1}$ is	B	18
R	The local maximum value of $f(x)=\frac{x^2+1}{x^2-1}, x\in R-\{-1,1\}$	C	-2
S	The local maximum value of $f(x)=\frac{x^2+1}{x}, x\in R-\{0\}$	D	2

1) P – C; Q – B; R – A; S – D **2)** P – D; Q – C; R – B; Q – A

3) P – B; Q – 1; R – A; S – C **4)** P – A; Q – D; R – C; S – B

19. $f(x)=x^2+\lambda x+\mu\cos x, \lambda\in\mathbb{Z}, \mu\in\mathbb{R}$. The number of ordered pair (λ,μ) for which $f(x)=0$ and $f(f(x))=0$ have same (non empty) set of real roots is

1) 4 **2) 6** **3) 8** **4) 10**

20. Suppose values taken by a variable X are such that $a\le x_i\le b$ where x_i denotes the value of X in the ith case for $i=1,2,.....,n.$ Then,

1) $\frac{a^2}{4}Var(X)$ **2)** $(b-a)^2\ge Var(X)$

3) $a\le Var(X)\le b$ **4)** $a^2\le Var(X)\le b^2$

SECTION-II (NUMERICAL VALUE TYPE)

This section contains **5 Numerical Value Type Questions**. The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases.

21. If $M(r)=\sin^r\theta+\cos^r\theta$, then Maximum value of $\left|\frac{M(2025)-M(2023)}{M(2021)}\right|(2024)$ is

Note: $|.|$ denotes absolute value of real number

22. If $a^2+b^2+c^2=1$, where $a,b,c\in\mathbb{R}$, then Maximum value of

$(4a-3b)^2+(5b-4c)^2+(3c-5a)^2$ is

23. Let $f:[2024, 2025]\to\mathbb{R}$ be a Differentiable function such that $f(2024)=2025$ and

$f(f(x))=x\forall x$ in [2024, 2025], then the value of $2025\int_{2024}^{2025}(x-f(x))^{2024}dx$ is

24. The sum of integers lie between 1 and 121 (both included) and divisible by 3 or 5 or 11 is_____

25. The area enclosed by the curve $xy^2=4(2-x)$ and y-axis is λ sq. unit, then the value of $\cos(1824\lambda)$ must be____.

PHYSICS

Max Marks: 100

SECTION-I (SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

26. A point moves in a straight line so that its displacement x (in m) at time t (in s) is given by $x^2 = 1 + t^2$. Its acceleration in m/s^2 at time t s is

1) $\frac{t}{x^3}$ **2)** $\frac{-t}{x^3}$ **3)** $\frac{1}{x} - \frac{t^2}{x^3}$ **4)** $\frac{1}{x} - \frac{1}{x^2}$

27. A solid sphere of radius R_1 and volume charge density $\rho = \frac{\rho_o}{r}$ is enclosed by a hollow sphere of radius R_2 with negative surface charge density σ, such that the total charge in the system is zero. ρ_o is a positive constant and r is the distance from the centre of the sphere. The ratio $\frac{R_2}{R_1}$ is

1) $\frac{\sigma}{\rho_o}$ **2)** $\sqrt{\sigma / \rho_o}$ **3)** $\sqrt{\rho_o / (2\sigma)}$ **4)** $\frac{\rho_o}{\sigma}$

28. A projectile is fired from level ground at an angle θ above the horizontal. The elevation angle ϕ of the highest point as seen from the launch point is related to θ by the relation.

1) $\tan\phi = \frac{1}{4}\tan\theta$ **2)** $\tan\phi = \tan\theta$ **3)** $\tan\phi = \frac{1}{2}\tan\theta$ **4)** $\tan\phi = 2\tan\theta$

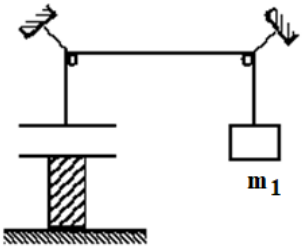
29. In a Searle's experiment, the diameter of the wire as measured by a screw gauge of least count 0.001cm is 0.050cm. The length measured by a scale of least count 0.1cm is 110.0cm. When a weight of 50N is suspended from the wire, the extension is measured to be 0.125cm by a micrometer of least count 0.001cm. The maximum percentage error in the measurement of Young's modulus of the material of the wire from this data is

1) 4.89% **2)** 3.89% **3)** 2.85% **4)** 2.75%

30. Water flows steadily through a horizontal pipe of variable cross-section. If the pressure of water is P at a point where flow speed is v, the pressure at another point where the flow speed is 2v, is (Take density of water as ρ)

1) $P - \frac{3\rho v^2}{2}$ **2)** $P - \frac{\rho v^2}{2}$ **3)** $P - \frac{3\rho v^2}{4}$ **4)** $P - \rho v^2$

31. In the given figure a capacitor of plate area A is charged upto charge q. The mass of each plate is m_2. The lower plate is rigidly fixed. The value of m_1 if the system remains in equilibrium is



1) $m_2 + \frac{q^2}{\in_0 Ag}$ **2)** m_2 **3)** $\frac{q^2}{2 \in_0 Ag} + m_2$ **4)** $2m_2$

32. The electron in a hydrogen atom makes a transition from an excited state to the ground state. Match the following.

a) Kinetic energy	p) increases
b) Potential energy	q) decreases
c) Total energy	r) remains the same

1) a-p, b-q, c-q **2)** a-p, b-q, c-q **3)** a-p, b-q, c-q **4)** a-p, b-q, c-q

33. **Assertion :** The ferromagnetic substance do not obey Curie's law as long it has ferromagnetic behavior.

Reason : At Curie point a ferromagnetic substance starts behaving as a paramagnetic substance and in this region susceptibility $\chi \propto \frac{1}{T - T_c}$.

1) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.

2) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.

3) If the Assertion is correct but Reason is incorrect.

4) If both the Assertion and Reason are incorrect.

34. STATEMENT-1: According to Archimedes' principle when body is floating, the weight of liquid displaced by body is equal to the weight of the body.

STATEMENT-2: A man is sitting in a boat which is floating on a pond. If the man drinks some water from the pond, the level of the water in the pond decreases

1) Both STATEMENT-1 and STATEMENT-2 are true

2) Both STATEMENT-1 and STATEMENT-2 are false.

3) STATEMENT-1 is true but STATEMENT-2 is false

4) STATEMENT-1 is false but STATEMENT-2 is true

35. If resistivity of the material of a conductor of uniform area of cross-section varies along its length as $\rho = \rho_0[1+\alpha x]$. Then the resistance of the conductor if its lengths is 'L' and area of cross-section is 'A' is

1) $\frac{\rho_0}{A}\left[L+\frac{1}{2}\alpha L^2\right]$ **2)** $\frac{\rho_0}{A}\left[L-\frac{1}{2}\alpha L^2\right]$ **3)** $\frac{\rho_0}{A}\left[L+\frac{1}{2}\alpha L\right]$ **4)** $\frac{\rho_0}{A}\left[L+\alpha L^2\right]$

36. A heavy weight is suspended from a spring. A person raises the weight till the spring becomes slack. The work done by him is W. The energy stored in the stretched spring was E. The gain in gravitational potential energy is

1) W **2)** E **3)** W + E **4)** W – E

37. A small candle 2.5 cm in size is placed at 27 cm in front of a concave mirror of radius of curvature 36 cm. At what distance from the mirror should a screen be placed in order to obtain a sharp image? What are the nature and size of image?

1) 54 cm, erected, magnified, 5cm

2) 54 cm, real, inverted, magnified, 5cm

3) 5 cm, erected, magnified, 54 cm

4) 54 cm, real, inverted, diminished, 4 cm

38. A uniform thin bar of mass 6m and length 12 L is bent to make a regular hexagon. Its moment of inertia about an axis passing through the centre of mass and perpendicular to the plane of the hexagon is

1) $20mL^2$ **2)** $6mL^2$ **3)** $\frac{12}{5}mL^2$ **4)** $30mL^2$

39. A beam of natural light falls on a system of 5 polaroids, which are arranged in succession such that the pass axis of each polaroid is turned through 60° with respect to the preceding one. The fraction of the incident light intensity that passes through the system is

1) $\frac{1}{64}$ **2)** $\frac{1}{512}$ **3)** $\frac{1}{256}$ **4)** $\frac{1}{128}$

40. A cubical block of mass M and edge *'a'* slides down a rough inclined plane of inclination θ with a uniform velocity. The torque of the normal force on the block about its centre has a magnitude.

1) Zero **2)** Mga **3)** $Mga\sin\theta$ **4)** $\frac{Mga\sin\theta}{2}$

41. A prism of refractive index n and angle A is placed in minimum deviation position. If the angle of minimum deviation is equal to the angle A, then the value of A is

1) $\sin^{-1}\left(\frac{n}{2}\right)$ **2)** $\sin^{-1}\left(\sqrt{\frac{n}{2}}\right)$ **3)** $2\sin^{-1}\left[\sqrt{\frac{1-n^2}{2}}\right]$ **4)** $2\sin^{-1}\left[\sqrt{\frac{4-n^2}{4}}\right]$

42. **Assertion:** The graph of total energy of a particle in SHM w.r.t., position is a straight line with zero slope.

Reason: Total energy of particle in SHM remains constant throughout its motion.

1) If both assertion and reason are true and the reason is the correct explanation of the assertion

2) If both assertion and reason are true but reason is not the correct explanation of the assertion.

3) If assertion is true but reason is false.

4) If the assertion and reason both are false.

43. A variable frequency a.c source is connected to a capacitor. With decrease in frequency, displacement current

1) Increases **2)** decreases **3)** remains same **4)** becomes zero

44. A light rod of length 200 cm is suspended from the ceiling horizontally by means of two vertical wires of equal length tied to its ends. One of the wires is made of steel and is of cross-section 0.1 cm^2 and the other of brass of cross-section 0.2 cm^2. Along the rod at which distance a weight may be hung to produce equal stresses in both the wires?

1) $\frac{4}{3}m$ from steel wire **2)** $\frac{4}{3}m$ from brass wire

3) 1 m from steel wire **4)** $\frac{1}{4}m$ from brass wire

45. **Assertion:** According to Huygen's principle, no backward wave-front is possible.

Reason: Amplitude of secondary wavelet is proportional to $(1 + \cos\theta)$ where θ is the angle between the ray at the point of consideration and the direction of secondary wavelet.

1) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.

2) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion

3) If the Assertion is correct but Reason is incorrect.

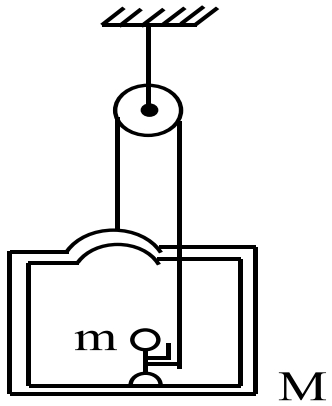
4) If both the Assertion and Reason are incorrect.

SECTION-II (NUMERICAL VALUE TYPE)

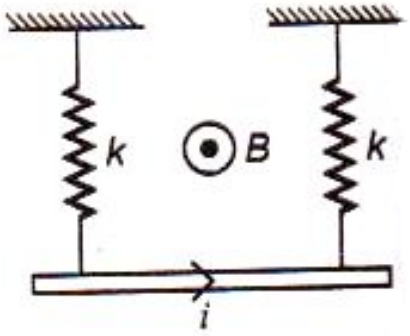
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46. A person of mass 50kg stands on a trolley of mass 30kg which is attached to a rope passing over a pulley. The person pulls down the other end of rope as shown in fig. such that trolley starts moving up with acceleration g/4. The force which person exerts on the rope in kg.wt is

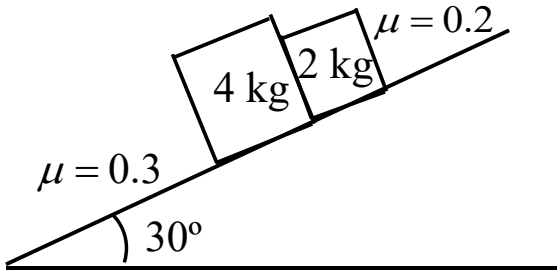


47. A current carrying conductor of mass 50g, length 0.5m carrying a current 1amp, hangs by two identical springs each of stiffness constant k = 50N/m. For an outward magnetic field B = 1 T, find extension(in cm) in the springs at equilibrium. ($g = 10 m/sec^2$)

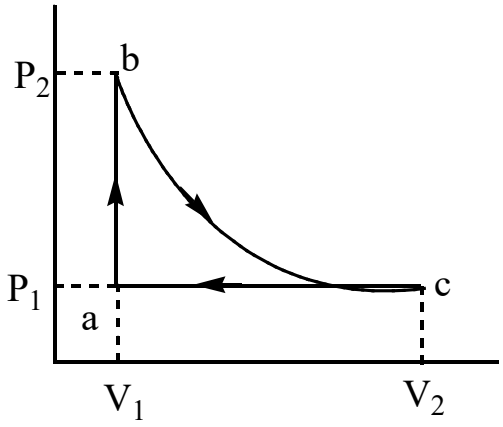


48. When a circular coil of radius 1 m and 100 turns is rotated in a horizontal uniform magnetic field, the peak value of emf induced is 100 V. The coil is unwound and then rewound into a circular coil of radius 2 m. If it is rotated, now, with the same speed under similar conditions, the new peak value of emf developed is (in V)

49. Two blocks 4 kg and 2 kg are sliding down an incline plane as shown in the figure. The acceleration of 2 kg block is



50. Carbon monoxide is carried around a closed cycle abca in which bc is an isothermal process, as shown in the figure. The gas absorbs 7000 J of heat, as its temperature increased from 300K to 1000K in going from a to b. The magnitude of heat ejected by the gas during the process ca is (in J)



CHEMISTRY

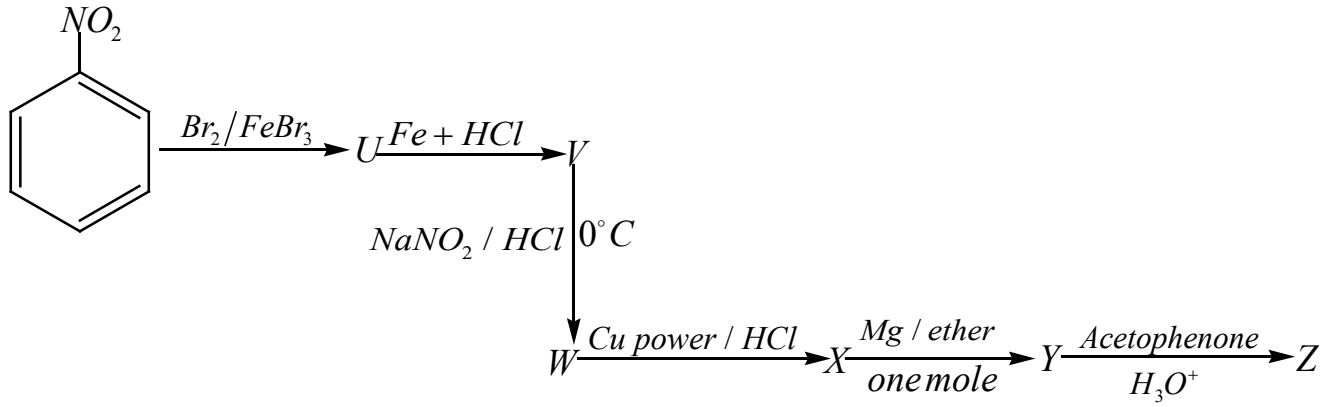
Max Marks: 100

SECTION-I (SINGLE CORRECT ANSWER TYPE)

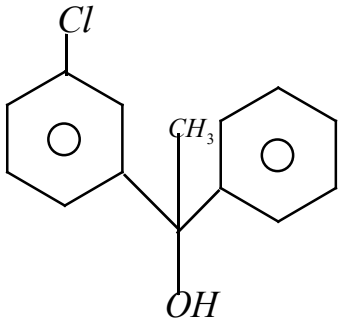
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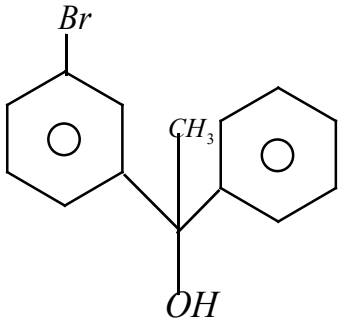
51. Identify the end product in the reaction sequence



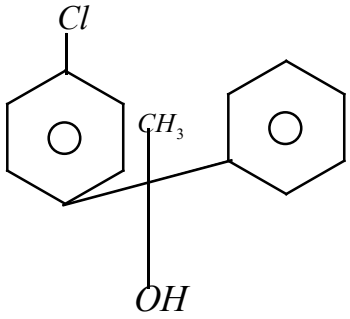
1)



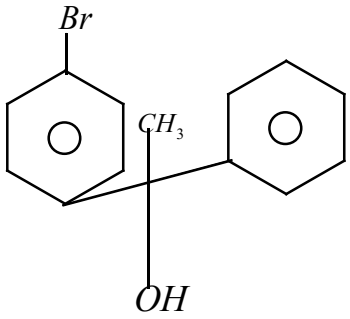
2)



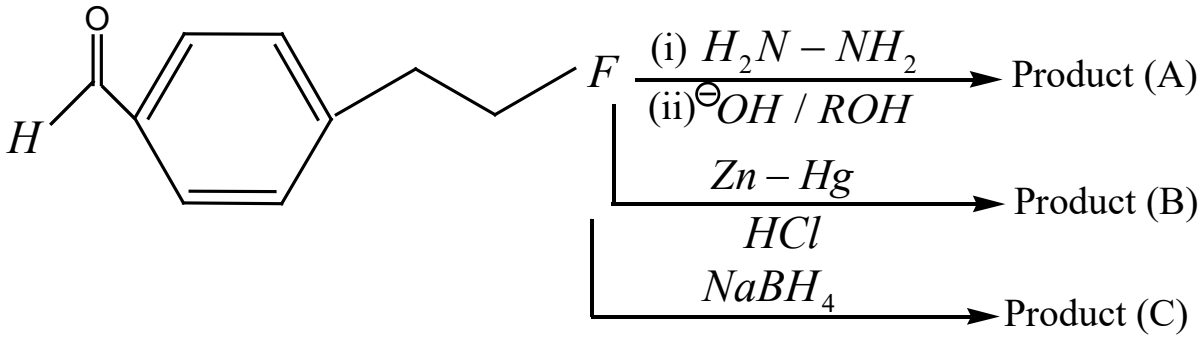
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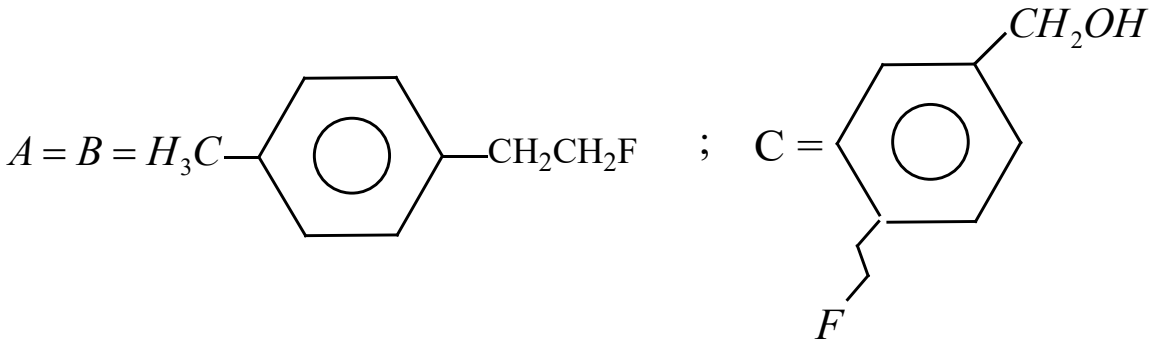
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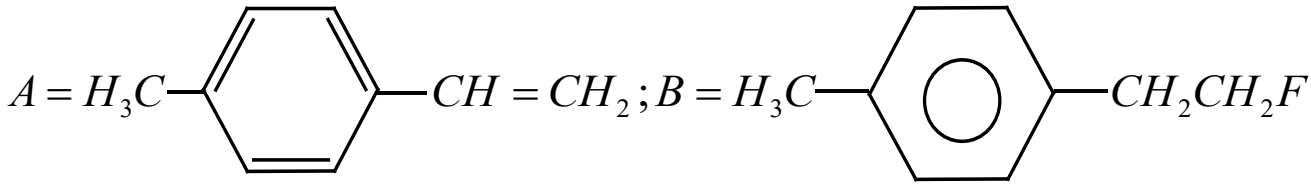


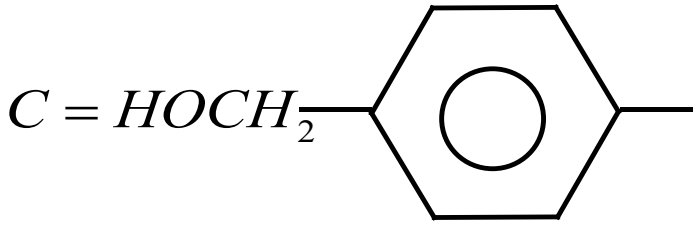
52.

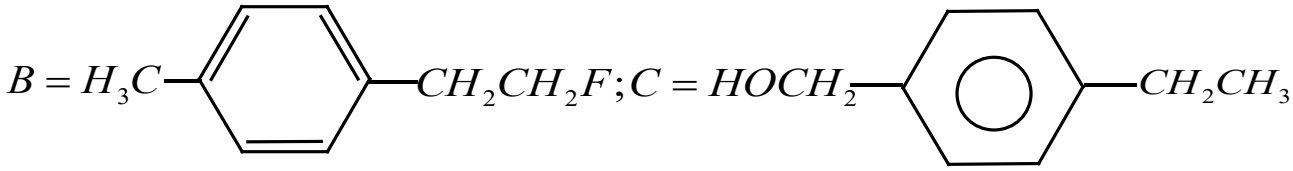


Correct option is

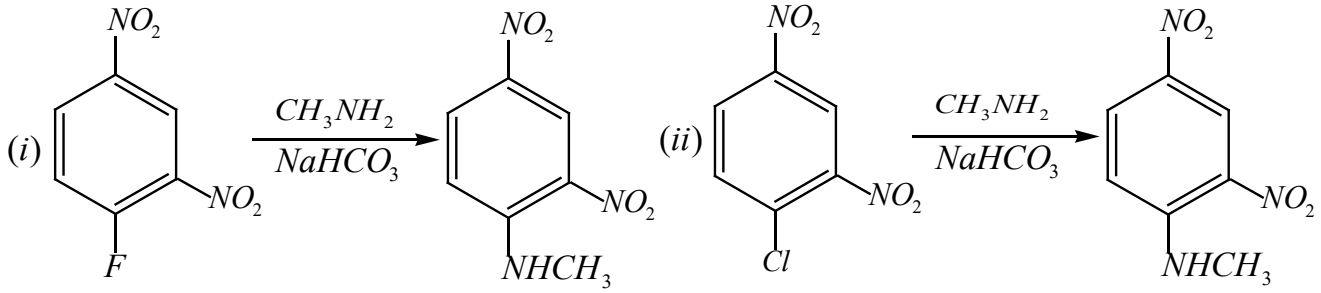
1) 

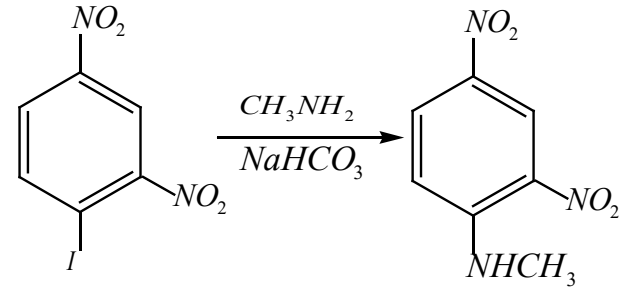
2) 

3) 

4) 

53. For the following reaction



(*iii*) 

Correct order of reactivity

1) i = ii = iii **2)** i > ii > iii **3)** i < ii < iii **4)** ii > iii > i

54. Assertion (A): It is difficult to replace chlorine by –OH in chlorobenzene in comparison to that in chloroethane.

Reason (R) : Chlorine – Carbon (C – Cl) bond in Chlorobenzene has a partial double bond character due to resonance.

1) Assertion (A) is true but Reason(R) is false

2) Both Assertion (A) and Reason(R) are true but Reason(R) is not the correct explanation of Assertion (A)

3) Assertion (A) is false but Reason(R) is true

4) Both Assertion (A) and Reason(R) are true and Reason(R) is the correct explanation of Assertion (A)

55. Which of the following statement is wrong?

1) Single N – N bond is stronger than the single P – P bond.

2) PH_3 can act as a ligand in the formation of coordination compound with transition elements.

3) NO_2 is paramagnetic in nature.

4) Covalency of nitrogen in N_2O_5 is four..

56. In qualitative analysis when H_2S is passed through an aqueous solution of salt acidified with HCl, a black precipitate is obtained. On boiling the precipitate with dil. HNO_3, it forms a solution of blue colour. Addition of excess of aqueous solution of ammonia to this solution gives

1) A deep blue precipitate of $Cu(OH)_2$

2) A deep blue solution of $\left[Cu(NH_3)_4\right]^{2+}$

3) A deep blue solution of $Cu(NO_3)_2$

4) A deep blue solution of $Cu(OH)_2.Cu(NH_3)_2$

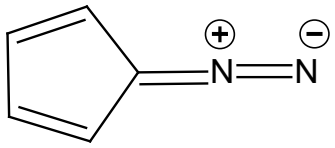
57. Match the complex ions given in Column I with the colours given in Column II and assign the correct code:

Column I(complex ion)		**Column II (colour)**	
A	$\left[Co(NH_3)_6\right]^{3+}$	P	Violet
B	$\left[Ti(H_2O)_6\right]^{3+}$	Q	Green

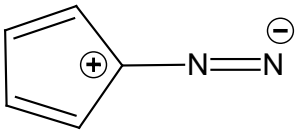
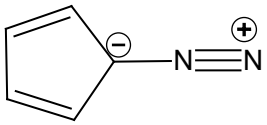
C	$\left[Ni\left(H_2O\right)_6\right]^{2+}$	S	Pale blue
D	$\left(Ni\left(H_2O\right)_4\left(en\right)\right]^{2+}\left(aq\right)\right)$	T	Yellowish orange
		U	Blue

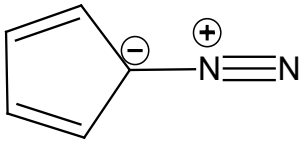
1) A (P), B (Q), C (T), D (U) **2)** A (T), B (S), C (Q), D (P)

3) A (S), B (Q), C (T), D (P) **4)** A (T), B (P), C (Q), D (S)



58. the most stable canonical structure of this molecule is

1)  **2)** 

3)  **4)** All are equally

59. The correct order of basicity of oxides of vanadium is

1) $V_2O_3 > V_2O_4 > V_2O_5$ **2)** $V_2O_3 > V_2O_5 > V_2O_4$

3) $V_2O_5 > V_2O_4 > V_2O_3$ **4)** $V_2O_4 > V_2O_3 > V_2O_5$

60. Strong reducing behavior of H_3PO_2 is due to

1) Low oxidation state of phosphorus

2) Presence of two -OH groups and one P – H bond

3) Two P – H bonds

4) High electron gain enthalpy of phosphorus

61. On addition of small amount of $KMnO_4$ to concentrated H_2SO_4, a green oily compound is obtained which is highly explosive in nature. Identify the compound from the following

1) Mn_2O_7 **2)** MnO_2 **3)** $MnSO_4$ **4)** Mn_2O_3

62. Which of the following statements are true?

1) $CH_3-CH_2-CH_2-I$ will react less readily than $\left(CH_3\right)_2CHI$ for S_N2 reactions

2) $CH_3-CH_2-CH_2-Cl$ will react more readily than $CH_3-CH_2-CH_2-Br$ for S_N2 reactions

3) $CH_3-CH_2-CH_2-CH_2-OH$ will react more readily than $\left(CH_3\right)_3C-CH_2-$OH for S_Ni reactions in the presence of pyridine

4) $CH_3-O-C_6H_5Br$ will react more readily than $NO_2-C_6H_5-CH_2Br$ for S_N2 reactions

63. Which of the following statement is not correct from the view point of molecular orbital theory?

1) Be_2 is not a stable molecule.

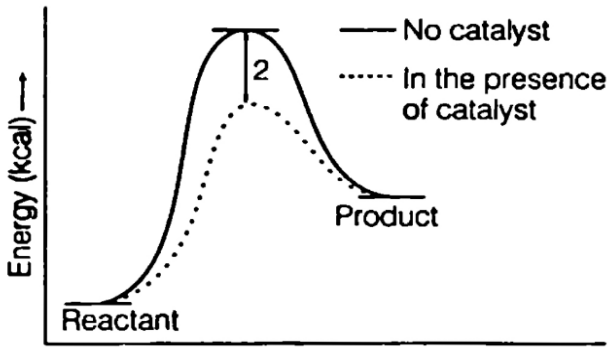
2) He_2 is not stable but He_2^+ is expected to exist.

3) Bond strength of N_2 is maximum amongst the homonuclear diatomic molecules belonging to the second period.

4) The order of energies of molecular orbitals in N_2 molecule is

$$\sigma 2s < \sigma^* 2s < \sigma 2p_z < \left(\pi 2p_x = \pi 2p_y\right) < \left(\pi^* 2p_x = \pi^* 2p_y\right) < \sigma^* 2p_z$$

64. How much rate is affected in the presence of catalyst at 300K? (find the approximate value)



1) 20 times **2)** 14 times **3)** 28 times **4)** 2 times

65. For the cell, $Zn\left|Zn^{2+}(1M)\right\|H^+(1M)\left|Pt(H_2), E^\circ_{cell} = 0.76V\right.$ and for the cell

$Pt(H_2)\left|H^+(1M)\right\|Ag^+(1M)\left|Ag\right.$, $E^\circ_{cell} = 0.80V$

Thus, E_{cell} for the

$Ag\left|Ag^+(0.1M)\right\|Zn^{2+}(0.1M)Zn$ is…….. and cell reaction is……

1) $1.44V$, spontaneous **2)** $0.4V$ spontaneous

3) $-1.44V$, non-spontaneous **4)** $-1.53V$, non-spontaneous

66. Given below are two statement

Statement-I: The gas liberated on warming a salt with dil H_2SO_4, turns a piece of paper dipped in lead acetate into black, It is a confirmatory test for sulphide ion.

Statement-II: In statement-I the colour of paper tunes black because of formation of lead sulphite.

In the light of the above statements, choose the most appropriate answer from the options below

1) Both statement-I and statement-II are false

2) Statement-I is false but statement-II is true

3) Statement-I is true but statement-II is false

4) Both statement-I and statement-II are true.

67. Statement-I: The enthalpy of neutralization of a strong acid with a strong base is always approximately $-57.1 kJ\,mol^{-1}$

Statement-II: The reaction between a strong acid and a strong base in an aqueous solution involves the formation of water molecules from H^+ and OH^- ions

1) Both statement-I and statement-II are correct, and statement-II is the correct explanation of statement-I.

2) Both statement-I and statement-II are correct, but statement-II is not the correct explanation of statement-I

3) Statement-I is correct, but statement-II is in correct

4) Statement-I is incorrect, but statement-II is correct

68. Assertion(A): The boiling point of a solution containing a non-volatile solute is always higher than that of the pure solvent

Reason(R): The addition of a non-volatile solute decreases the vapor pressure of the solution.

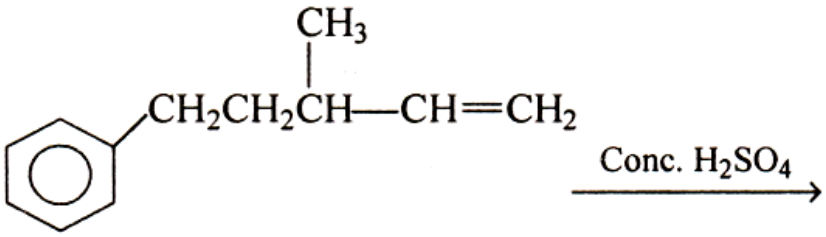
1) Both Assertion (A) and Reason(R) are correct, and Reason(R) is the correct explanation of Assertion (A)

2) Both Assertion (A) and Reason(R) are correct, but Reason(R) is not the correct explanation of Assertion (A)

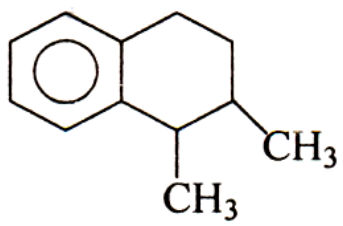
3) Assertion (A)is correct , but Reason(R) is incorrect

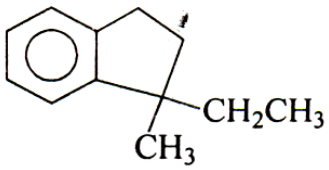
4) Assertion (A)is incorrect, but Reason(R) is correct

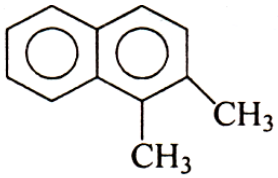
69. Consider the following reaction

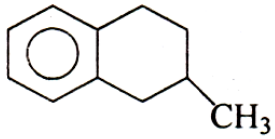


The major product is expected to be :

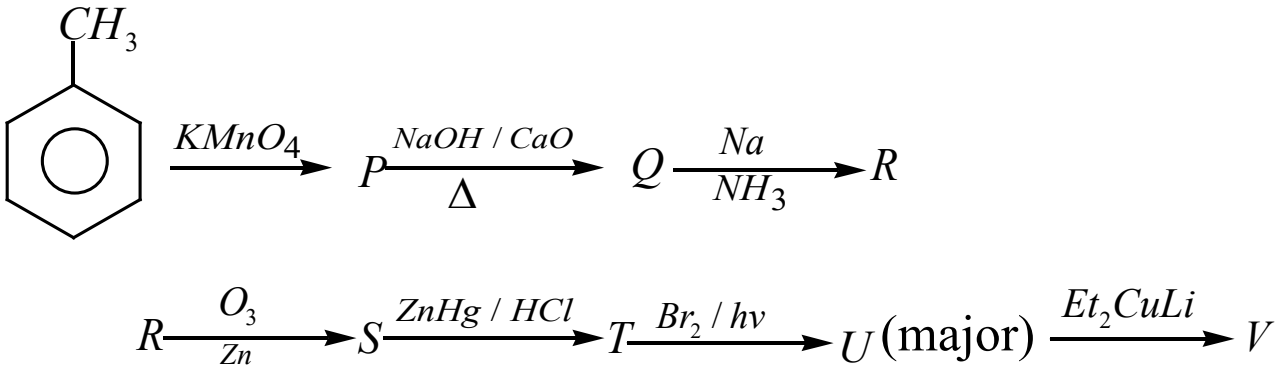
1) 

2) 

3) 

4) 

70.



The number of monochloro products of compound 'V' and the fractional distillation of product mixture respectively are:

1) 4, 4 **2)** 6, 5 **3)** 6, 4 **4)** 4, 2

SECTION-II (NUMERICAL VALUE TYPE)

This section contains **5 Numerical Value Type Questions**. The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

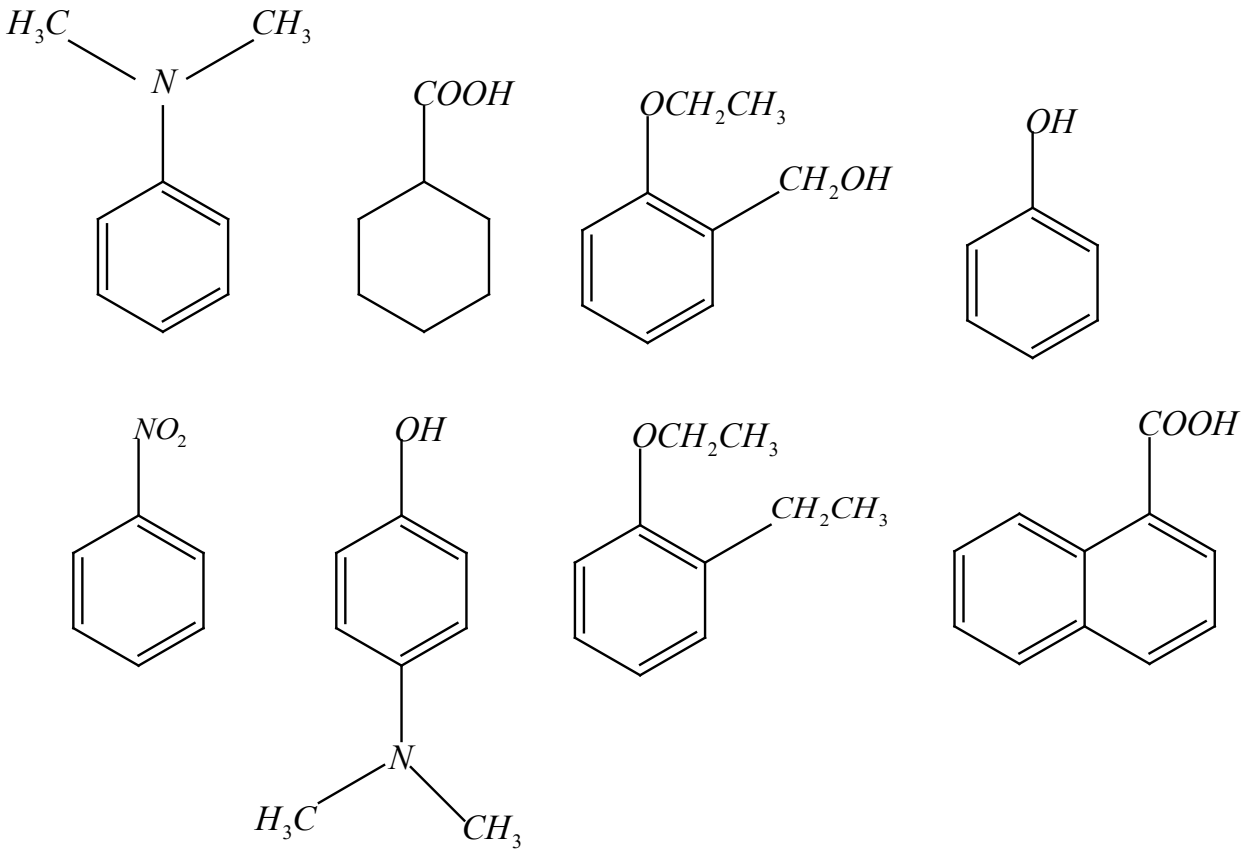
Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases

71. Adenine, a nitrogenous base found in DNA and RNA, the % of N in adenine is X. The value of X is (round off nearest integer)

Given: Molar mass

$N = 14\,g\,mol^{-1}; O = 16\,g\,mol^{-1}; C = 12\,g\,mol^{-1}; H = 1\,g\,mol^{-1}$

72. The number of compounds from the following, soluble in aqueous $NaOH$ is



73. 1 g of a monobasic acid dissolved in 200 g of water lowers the FP by $0.186^{\circ}C$. On the other hand when 1 g of the same acid is dissolved in water so as to make the solution $200\,mL$, this solution requires $125\,mL$ of $0.1N\ NaOH$ for complete neutralization. Calculate α, is the degree of dissociation of acid . $\left(K_{f(H_2O)} = 1.86\ \frac{K-kg}{mol}\right)$, find the value of 10α

74. The molar solubility of $Fe(OH)_3$ in a buffer solution that 0.10 M in NH_4Cl and that 0.10 M in NH_3 is $a \times 10^{-27} M$. The value of **a is -----**

Given: $K_b\left(NH_3\right) = 1.8 \times 10^{-5}$ and $K_{sp}\left[Fe\left(OH\right)_3\right] = 2.6 \times 10^{-39}$.

75. Among the following, the number of planar as well as polar species is:

$XeO_3, ClF_3, SO_3, HNO_3, SF_4, BeF_2, SO_2, XeF_4, HCl$